From Classroom/Integrated School Programs 7 in Top 20, 23 in Top 100, 54 in Top 300, 106 in Top 500 All India Ranks & 2314 Students from Classroom /Integrated School Programs & 3723 Students from All Programs have been Awarded a Rank in JEE (Advanced), 2013

ANSWERS, HINTS & SOLUTIONS

FULL TEST –I
(Paper-2)

Q. No.	PHYSICS	CHEMISTRY	MATHEMATICS
1.	C	D	B
2.	C	A	B
3.	D	C	A
4.	B	B	C
5.	B	B	D
6.	D	C	D
7.	B	D	C
8.	C	B	A
9.	A	C	C
10.	D	B	C
11.	B	A	D
12.	D	C	A
13.	B	A	C
14.	A	D	A
15.	A	C	A
16.	B	B	C
17.	B	A	B
18.	A	B	A
19.	B	C	A
1.	(A) → (s), (B) → (q), (C) → (p), (D) → (r)	(A → p, q, r), (B → q), (C → p, r, S), (D → p)	(A) → (q), (B) → (r), (C) → (s), (D) → (p)
2.	(A) → (q); (B) → (r); (C) → (s); (D) → (p)	(A → p, q, s), (B → r, s), (C → q), (D → r, s)	(A) → (s), (B) → (p), (C) → (q), (D) → (r)
3.	(A) → (p), (B) → (r), (C) → (s), (D) → (q)	(A → p, r), (B → q, p), (C → q), (D → q, s)	(A) → (s), (B) → (r), (C) → (p), (D) → (q)

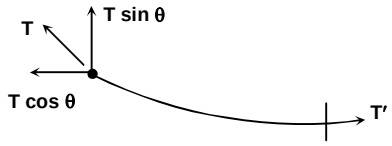
Physics

PART – I

2. $T \sin\theta = mg/2$

$T' = T\cos\theta$

$= \frac{mg}{2}\cot\theta = \frac{mg}{2\tan\theta}$



3. $\frac{1}{2}mv^2 = pt$ (P = const)

$\Rightarrow \quad v = \sqrt{\frac{2Pt}{m}}$

$a = \frac{dv}{dt} = \sqrt{\frac{2P}{m}}\frac{1}{2\sqrt{t}}$

$F = ma = \sqrt{\frac{mP}{2t}} \propto \frac{1}{v}$

5. $\vec{F} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4 + \vec{F}_5$

$\left|\vec{F}_2\right| = \left|\vec{F}_5\right|$ and $\left|\vec{F}_2\right| = \left|\vec{F}_4\right|$

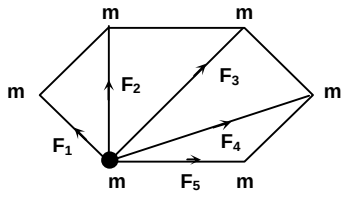
$\vec{F}_1 = \vec{F}_3 + 2F_2\cos 30° + 2F_1\cos 60°$

$F_3 = \frac{Gm^2}{4a^2}$; $F_2 = \frac{Gm^2}{3a^2}$; $F_1 = \frac{Gm^2}{a^2}$

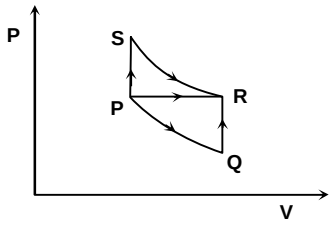
$F = \frac{Gm^2}{a^2}\left(\frac{5}{4} + \frac{1}{\sqrt{3}}\right) = m\omega^2 a$

$\omega = \sqrt{\frac{Gm}{a^3}\left(\frac{5}{4} + \frac{1}{\sqrt{3}}\right)}$

$T = 2\pi\sqrt{\frac{4\sqrt{3}a^3}{Gm\left(5\sqrt{3}+4\right)}}$



6.



7. $a = \frac{\rho v(g+a) - m(g+a)}{m} = 20\ m/s^2$

$t = \sqrt{\frac{2h}{a}} = 1\ sec$

10. Time period becomes $2\pi\sqrt{\frac{R}{g}}$. We can't neglect the roundness of earth for the pendulum of infinite length.

12. The –ve sign is $-\frac{d\phi}{dt}$ signifies the direction of induced emf.

14. process AB $U\rho$ = constant

$\frac{P}{\rho}=\frac{RT}{M}$ and $U \propto t$

$\Rightarrow$ P = const

Process BC $\rightarrow$isochoric

Process CA $\rightarrow$ isothermal

15. $Q = Q_{AB} + Q_{BC} + Q_{CA}$

$Q = -5U_0 + 3U_0 + \frac{10U_0}{3}\ln 2.5$

16. $W_{AB} = \Delta Q_{AB} - \Delta U_{AB} = -5U_0 - (-3U_0) = -2U_0$

17. For lens L_1, ray must move parallel to the axis after refraction $\frac{\mu_1}{\infty}+\frac{\mu_w}{x}=\frac{\mu_1-\mu_\omega}{R_1} \Rightarrow x = 10$ cm

18. For lens L_2, image must form at centre of curvature of the curved surface after refraction through plane part.

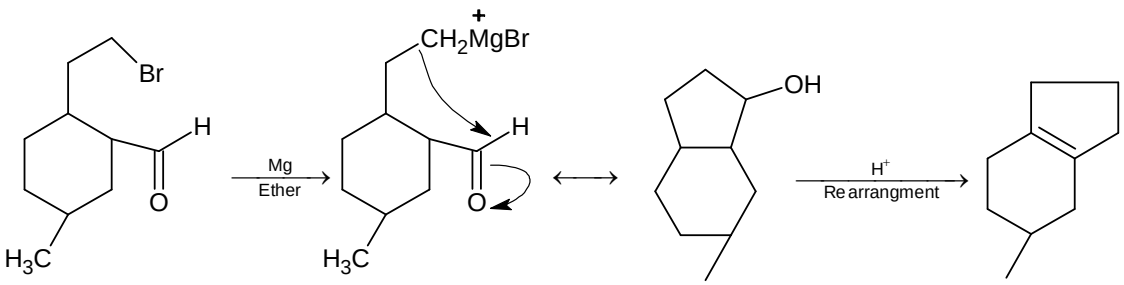
$\frac{\mu_2}{-R_2}+\frac{\mu_\omega}{x'}=0$

$\Rightarrow$ $x' = 8$ cm

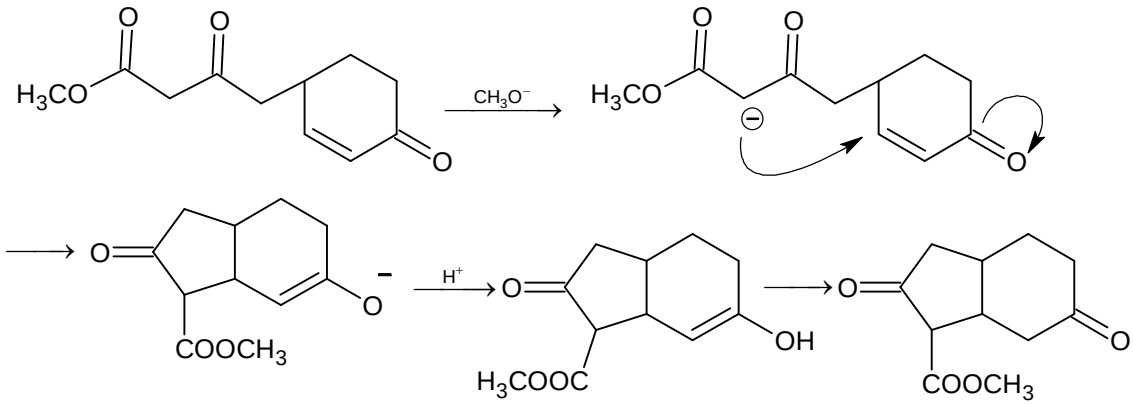
Chemistry

PART – II

1.



2.



3. $P = P_A^o X_A + P_B^o = P_A^o (1 - X_B) + P_B^o X_B$

$\Rightarrow P = P_A^o - \left(P_A^o - P_B^o\right) \times B$

Thus $P_A^o = 120$ Torr

$P_A^o - P_B^o = 75 \Rightarrow P_B^o = 45$ Torr

Hence C is correct answer.

4. $\Lambda^o_{BaSO_4} = \kappa \times \dfrac{1000}{\text{Conc(normality)}}$

$\text{Normality} = \dfrac{\kappa \times 1000}{\Lambda^o_{BaSO_4}} = \dfrac{1000 \times 8 \times 10^{-5}}{400} = 2 \times 10^{-4}.$

$\text{Molarity} = \dfrac{\text{Normality}}{2} = 10^{-4} M\,(\text{Solubility})$

$Ksp = S^2 = 10^{-8} M^2.$

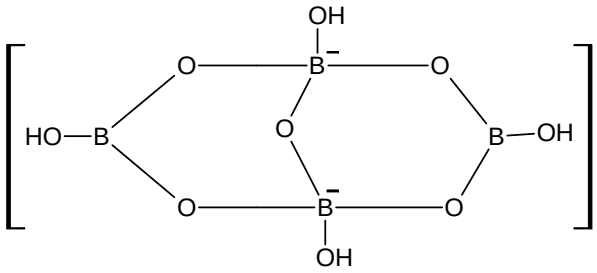
5. $T_{av} = \dfrac{1}{K}$

$T_{99.9} = 10 \times ty_2 = \dfrac{10 \times 0.693}{K} = \dfrac{6.93}{K}$

Number of natural life times = $\left(\dfrac{6.693}{K}\right) / \left(\dfrac{1}{K}\right)$

= 6.93

6.



7. $\underset{145}{Ca(OCl)Cl.H_2O} \longrightarrow \underset{71}{Cl_2}$

Percentage $= \frac{71}{145} \times 100 = 49$

Hence D is correct answer.

17. $\left[\frac{d(c)}{dt}\right]_I = \frac{0.0033}{25} = 1.32 \times 10^{-4} \text{m min}^{-1}$

$\left(\frac{d(c)}{dt}\right)_{II} = 2.6 \times 10^{-4} \text{ m min}^{-1}$

$\left(\frac{d(c)}{dt}\right)_{III} = 1.02 \times 10^{-3} \text{m min}^{-1}$

18. On comparing rates order w.r.t A = 2, and w.r.t. B = 1. Thus rate law = $K[A]^2[B]$

19. $\frac{dx}{dt} = K[A]^2[B]$

$K = \frac{(dx/dt)}{[A]^2[B]} = 0.26$

Mathematics

PART – III

1. Total number of lines made = $^9C_2 = 36$
Now, these 36 lines are 9 sets each with 4 parallel members. With 5 vertices number of lines made
$= {}^5C_2 = 10$
Clearly, atleast 2 members belong to the same set
So, atleast one pair is parallel

2. $\cos\theta = \dfrac{OA^2 + OB^2 - AB^2}{2OA \cdot OB}$

$= \dfrac{OA^2 + OB^2 - \left(\dfrac{OA + OB}{2}\right)^2}{2OA \cdot OB} = \dfrac{3}{8}\dfrac{(OA^2 + OB^2)}{OA \cdot OB} - \dfrac{1}{4}$

For maximum $\cos\theta$,

$\dfrac{3}{8}\left(\dfrac{OA^2 + OB^2}{OA \cdot OB}\right) - \dfrac{1}{2} \geq \dfrac{3}{8} \times \dfrac{2 \cdot OA \cdot OB}{OA \cdot OB} - \dfrac{1}{4} = \dfrac{1}{2}$

$\Rightarrow \theta = \dfrac{\pi}{3}$

3. $S_1 = a_1$
$S_2 = a_1 + a_2$
$S_3 = a_1 + a_2 + a_3$
$\vdots$
$S_n = a_1 + a_2 + \ldots\ldots + a_n$
If we divide all $S_1, S_2, \ldots\ldots S_n$ by 23 we get remainders 1, 2, 3, 4, ..., 22
So, two of these give same remainders S_p, S_q
$\Rightarrow S_p - S_q$ will be divisible by 23
$S_p - S_q = a_{p+1} + a_{p+2} + \ldots\ldots + a_q$

4. Using Cauchy we get, $\left(|z_1| + |z_2| + |z_3| + |z_4|\right)\left(\dfrac{1^2}{|z_1|} + \dfrac{1^2}{|z_2|} + \dfrac{2^2}{|z_3|} + \dfrac{4^2}{|z_4|}\right) \geq (1+1+2+4)^2 = 64$

5. Point of intersection is $\theta = \dfrac{1}{2}\ln\left(\dfrac{b}{a}\right)$

For C_1,

$\dfrac{dr}{d\theta} = ae^{\theta}$, now $\tan\phi_1 = r\dfrac{d\theta}{dr} = (ae^{\theta})a^{-1}e^{-\theta} = 1$

$\therefore \phi_1 = \dfrac{\pi}{4}$

For C_2,

$\dfrac{dr}{d\theta} = -be^{-\theta} \Rightarrow \tan\phi_2 = (be^{-\theta})\left(-\dfrac{1}{b}\right)e^{\theta} = -1$

$\therefore \phi_2 = \dfrac{3\pi}{4}$

$\therefore$ Angle of intersection is $\phi_2 - \phi_1 = \dfrac{3\pi}{4} - \dfrac{\pi}{4} = \dfrac{\pi}{2}$

6. Let c, d, e be the three points where y = f(x) crosses x–axis. Then, f(c) = f(d) = f(e) = 0
Assuming a < c < d < e < b. The function f satisfies Rolle's theorem in two intervals (c, d) and (d, e). Since f and f′ are continuous and f(c) = f(d) = 0
So, there exists, at least one point in the interval (c, d) and (d, e) such that derivative is zero
Let, $C_1 \in$ (c, d) such that $f'(C_1) = 0$ and $C_2 \in$ (d, e) such that $f'(C_2) = 0$. Now the function f′ satisfies Rolle's theorem since f′, f″ are continuous and $f'(C_1) = f'(C_2) = 0$
So, by Rolle's theorem, there exists a number C_3 in between C_1 and C_2 such that $f''(C_3) = 0$
$\Rightarrow$ Minimum one root C_3 of the equation $f''(x) = 0$ lies in the interval (a, b)

7. Dividing the given differential equation by $3xy(y^2 - x^2)$

$$\frac{y\left(y^2-2x^2\right)}{3xy\left(y^2-x^2\right)}dx+\frac{x\left(2y^2-x^2\right)}{3xy\left(y^2-x^2\right)}dy=0$$

$$\Rightarrow \frac{dx}{x}-\frac{xdx}{y^2-x^2}+\frac{ydy}{y^2-x^2}+\frac{dy}{y}=0$$

$$\Rightarrow d\left(\ln(xy)\right)+\frac{1}{2}\frac{d\left(y^2-x^2\right)}{y^2-x^2}=0$$

$$\Rightarrow d\left[\ln\left[x^2y^2\left(y^2-x^2\right)\right]\right]=0$$

$$\Rightarrow \ln\left[x^2y^2\left(y^2-x^2\right)\right]=c$$

$\Rightarrow x^2y^2(y^2 - x^2) = c$

8. Let the circle be $x^2+(y-\alpha)^2=a^2$. Let the point of intersection of tangents at P and Q be (h, k).
Then equation of PQ, is $hx+(k-\alpha)(y-\alpha)-a^2=0$. As it passes through $(a,0)$, so,
$ha-\alpha(k-\alpha)-a^2=0$.
$\Rightarrow \alpha^2-k\alpha+a(h-a)=0.\ D\geq 0 \Rightarrow k^2-4a(h-a)\geq 0$
i.e. $y^2\geq 4a(x-a)$.

9. Consider $\int_0^1(\alpha-x)^2 f(x)dx=\int_0^1\left(\alpha^2f(x)-2\alpha xf(x)+x^2f(x)\right)dx = \alpha^2 - 2\alpha^2 + \alpha^2 = 0$
However f(x) assumes only positive values i.e. in (0, 1)
$\Rightarrow (\alpha - x)^2 (f(x)) > 0 \Rightarrow$ integral can't be zero

10. Differential equation can be written as, (p – x)(p – 2 sin x)(2p + cos x) = 0 which has solution as $(2y - x^2 - c)(y + 2\cos x - c)(2y + \sin x - c) = 0$

11. Put x = –1 we get $(-1 + 1)\, p(-1) + 1 = \alpha(-1)^{n+1}(n + 1)!$

$$\alpha=\frac{(-1)^{n+1}}{(n+1)!}$$

So, $p(x)=\frac{(-1)^{n+1}x(x-1).....(x-n)}{(n+1)!(n+1)}+\frac{x}{x+1}$

Clearly, $p(n+1)=\begin{cases}1, & \text{where 'n' is odd}\\ \frac{n}{n+2}, & \text{where 'n' is even}\end{cases}$

13. If e is the eccentricity then, $e^2 = \frac{\alpha^2 - \beta^2}{\alpha^2}$

Now, we know, $\frac{1}{\alpha^2} + \frac{1}{\beta^2} = a + b \Rightarrow \frac{1}{\alpha^2\beta^2} = ab - h^2$

$\Rightarrow \alpha^2 + \beta^2 = \frac{a+b}{ab-h^2}$, $\alpha^2\beta^2 = \frac{1}{ab-h^2}$

$\therefore \alpha^2 - \beta^2 = \sqrt{\frac{(a+b)^2 - 4(ab-h^2)}{(ab-h)^2}}$

$\frac{1}{\alpha^2} - \frac{1}{\beta^2} = \mp\sqrt{(a+b)^2 - 4(ab-h^2)}$

For an ellipse $\frac{2}{\alpha^2} = a + b - \sqrt{(a+b)^2 - 4(ab-h^2)}$

So, $e^2 = \frac{\sqrt{(a-b)^2 + 4h^2}}{2(ab-h^2)}\left[(a+b) - \sqrt{(a-b)^2 + 4h^2}\right]$

14. Put y = z = t = 0
f(0)[f(x) + f(0)] = f(0)
Put x = 0
$2f^2(0) = f(0)$
$\Rightarrow f(0) = 0, \frac{1}{2}$
If $f(0) = \frac{1}{2} \Rightarrow f(x) + \frac{1}{2} = 1$
$\Rightarrow f(x) = \frac{1}{2}$
If f(0) = 0, z = t = 0
$\Rightarrow$ f(x) f(y) = f(xy)
Let, x = y = 1 $\Rightarrow f^2(1) = f(1)$
$\Rightarrow$ f(1) = 0 or f(1) = 1
We have f(0) = 0, f(1) = 0, y = 1
f(x) = 0
Also, f(0) = 0, f(1) = 1, x = 0, y = t = 1
(f(0) + f(z)) (f(1) + f(1)) = f(–z) + f(z)
$\Rightarrow$ 2f(z) = f(–z) + f(z)
$\Rightarrow$ f(z) = f(–z)

15. If y = x in f(x) f(y) = f(xy)
$\Rightarrow f(x^2) = f^2(x) \geq 0$
Put x = t, y = z
$[f(x) + f(y)]^2 = f(x^2 + y^2)$
$\Rightarrow f(x^2 + y^2) = f^2(x) + f^2(y) + 2f(x)f(y) \geq f^2(x)$
$\Rightarrow f(x^2 + y^2) \geq f(x^2)$
$\Rightarrow$ f is non decreasing for positive x

16. Put y = z = t = 1
$\Rightarrow$ 2(f(x) + 1) = f(x – 1) + f(x + 1)
f(2) = 4, f(z) = 9, f(1) = 1, f(0) = 0
$f(n) = n^2$ (Possible function), if $f(n - 1) = (n - 1)^2$
2[f(n – 1) + 1] = f(n – 2) + f(n)

$\Rightarrow f(n) = n^2$

Now, for $x = \frac{p}{q}$ (rational number)

$f\left(\frac{p}{q}\right) f\left(q^2\right) = f(pq) \Rightarrow f\left(\frac{p}{q}\right) q^2 = p^2 q^2$

$\Rightarrow f\left(\frac{p}{q}\right) = \left(\frac{p}{q}\right)^2$ (True for rational number)

Now, if $x \in R$, lets prove for positive x since if it is proved the function is even and will follow for negative x

Assume for $x > 0$, $f(x) < x^2$

So, now a rational number 'r' between $\sqrt{f(x)}$ and $x \Rightarrow \sqrt{f(x)} < r < x$

$\Rightarrow f(x) < r^2 < x^2$

[$f(r) = r^2$, f is non decreasing]

$\Rightarrow f(r) = r^2 \le f(x)$ [contradiction]

$\Rightarrow f(x) < x^2$ (impossible)

Similarly we can prove contradiction $f(x) > x^2$

So, only possibility $f(x) = x^2$ substituting $f(x) = x^2$ we get

$(f(x) + f(z))\,(f(y) + f(t)) = f(xy - zt) + f(xt + yz)$

$(x^2 + z^2)(y^2 + t^2) = (xy - zt)^2 + (xt + yz)^2$ [Lagrange identity]

So, $f(x) = \frac{1}{2}$, $f(x) = 0$, $f(x) = x^2$ are the required solutions

17. $am^3 + m(2a - h) + k = 0$

$\Rightarrow am^3 + m(2a - x_1) = 0$

$\Rightarrow am^2 = x_1 - 2a$ {m = 0 (one possible value)}

$\Rightarrow m^2 = \frac{x_1 - 2a}{a}$, if $x \in (0, 2a)$

$\Rightarrow m^2$ = (–) number, so non real roots

18. Let P(h, k) be the point $am^3 + m(2a - h) + k = 0$

Since, m_1, m_1, m_1 are the possible roots

$\Rightarrow 3m_1 = 0 \Rightarrow m_1 = 0$

If $m_1 = 0$ is the root then k = 0

$am^3 + m(2a - h) = 0$

$\Rightarrow m^2 = \frac{h - 2a}{a} = 0 \Rightarrow h = 2a$

(2a, 0) is the only point

19. $A_1 = \int_{2a}^{8a} \sqrt{\frac{4}{27a}}(x - 2a)^{3/2}\,dx = \frac{2}{3\sqrt{3a}} \frac{2}{5}\left[(x - 2a)^{5/2}\right]_{2a}^{8a}$

$= \frac{2}{3\sqrt{3a}} \frac{2}{5}(6a)^{5/2} = \frac{2}{3\sqrt{3a}} \times \frac{2}{5} \times 36a^2\sqrt{6a} = \frac{48\sqrt{2}a^2}{5}$

$A_1 = \int_0^{8a} \sqrt{4ax}\,dx = 2\sqrt{a} \times \frac{2}{3}\left[x^{3/2}\right]_0^{8a} = \frac{4}{3}\sqrt{a}\ 8a\sqrt{8a} = \frac{64}{3}\sqrt{2}a^2$

Area = $2(A_2 - A_1) = 2 \times 16\sqrt{2}a^2\left[\frac{4}{3} - \frac{3}{5}\right] = 32\sqrt{2}a^2 \times \frac{11}{15} = \frac{352\sqrt{2}a^2}{15}$

SECTION – B

1. **(A)** $|A - xI| = 0$

There exists a non zero matrix X such that $AX = xX$

$\Rightarrow [(\text{Adj } A)A]X = x(\text{adj } A)X$

$\Rightarrow |A|IX = x(\text{adj } A)X$

$\Rightarrow |A|X = x(\text{adj } A)X$

$\Rightarrow \dfrac{|A|X}{x} = (\text{adj } A)X$

$\Rightarrow (\text{adj } A)X = \dfrac{|A|X}{2}$

(B) $|\bar{A}' - \bar{\lambda}I| = |\overline{(A - \lambda I)'}| = |\overline{A - \lambda I}|$

$\therefore |\bar{A}' - \bar{\lambda}I| = 0$ if and only if $|\overline{A - \lambda I}| = 0$

$\Rightarrow |\bar{A}' - \bar{\lambda}I| = 0$ if and only if $|\overline{A - \lambda I}| = 0$

or $\bar{\lambda}$ is the root of $|\bar{A}' - xI| = 0$ if and only if λ is the root of $|A - yI| = 0$

$\therefore e^{-i\alpha}$ is the required solution

(C) $|B - \lambda I| = \begin{bmatrix} |A_{11}| - \lambda & |A_{12}| ..... & |A_{1n}| \\ 0 & |A_{22}| - \lambda ..... & |A_{2n}| \\ 0 & | & | \\ | & | & | \\ | & | & | \\ | & | & |A_{nn}| - \lambda \end{bmatrix} = 0$

$\Rightarrow (|A_{11}| - \lambda)(|A_{22}| - \lambda) \ldots\ldots (|A_{nn}| - \lambda) - 0$

Clearly the elements of principal diagonal become the roots

(D) Let, $AX = \lambda X$ [X is a non zero matrix]

$\Rightarrow \bar{X}'AX = \bar{X}'\lambda X = \lambda \bar{X}' = A\bar{X}'IX$

$\bar{X}'AX$ and $\bar{X}'IX$ are both real

Also, $\bar{X}'X \neq 0$, $X \neq 0$

$\Rightarrow \lambda = \bar{X}'AX / \bar{X}IX$ is real so λ can have real values

2. **(A)** $z = \cos\theta + i\sin\theta$, $z + \dfrac{1}{z} = 2\cos\theta$

$z - \dfrac{1}{z} = 2i\sin\theta$

$z^p + \dfrac{1}{z^p} = 2\cos p\theta$

Now, $(2i\sin\theta)^4 (2\cos\theta)^2 = \left(z - \dfrac{1}{z}\right)^4 \left(z + \dfrac{1}{z}\right)^2 = \left(z^6 + \dfrac{1}{z^6}\right) - 2\left(z^4 + \dfrac{1}{z^4}\right) - \left(z^2 + \dfrac{1}{z^2}\right) + 4$

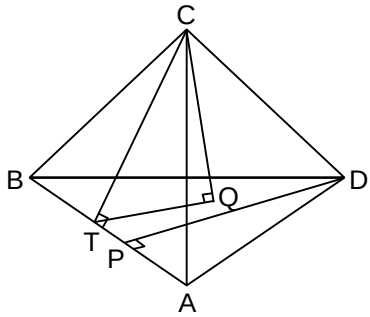
then, $2\cos 6\theta - 2.2.\cos 4\theta - 2\cos 2\theta + 4$

$\Rightarrow 2^4 i^4 \sin^4\theta\, 2^2 \cos^2\theta = 2(\cos 6\theta - 2\cos 4\theta - \cos 2\theta + 2)$

$\Rightarrow \lambda = 2$

(B) Volume of tetrahedron = $\frac{1}{3}\times(\text{Ar}\Delta ABD)\times CQ$

$= \frac{1}{3}\left(\frac{AB}{2}\right)\times DP\times CQ$



Let, CD be the largest side and AB = $x \le 1$
Let, T be closer to A than B

$\Rightarrow BT \ge \frac{x}{2}$

$\Rightarrow CT^2 = BC^2 - BT^2 \le 1-\frac{x^2}{4}$

$\Rightarrow CT \le \sqrt{1-\frac{x^2}{4}}$ (same for DP, CQ)

So, $V \le \frac{1}{3}\times\frac{x}{2}\times\left(1-\frac{x^2}{4}\right) = \frac{1}{24}x(4-x^2) \Rightarrow V_{max} = \frac{1}{8}$

(C) By AM $\ge$ GM

$\cos^3 x + \frac{\cos x}{4} \ge \cos^2 x$ (x is an acute angle)

Now, setting x = A, B, C we get
$x_1 + x_3 \ge \cos^2 A + \cos^2 B + \cos^2 C + 2\cos A\cos B\cos C = 2x_2$
$\Rightarrow x_1 + x_2 + x_3 \ge 3x_2 = \frac{3}{2}$

(D) Let, $u = \cot\frac{A}{2}$, $v = \cot\frac{B}{2}$, $w = \cot\frac{C}{2} \Rightarrow u = \frac{S-a}{r}$, $v = \frac{S-b}{r}$, $w = \frac{S-c}{r}$

$\frac{S}{r} = \frac{(S-a)+(S-b)+(S-c)}{r} = u+v+w$

Now we can rewrite as $49[u^2 + 4v^2 + 9w^2] = 36(u+v+w)^2$
$\Rightarrow (3u-12v)^2 + (4v-9w)^2 + (18w-2u)^2 = 0$
$\therefore u : v : w = 1 : \frac{1}{4} : \frac{1}{9}$

Multiplying by r we get $\frac{S-a}{36} = \frac{S-b}{9} = \frac{S-c}{4} = \frac{2S-b-c}{9+4}$

$= \frac{2S-c-a}{4+36} = \frac{2S-a-b}{36+9} \Rightarrow \frac{a}{13} = \frac{b}{40} = \frac{c}{45}$

3. **(A)** Put $x = y + \frac{1}{2}$ and substitute, $8y^4 + 4y^2 + a - \frac{3}{2} = 0$

Again, $z = y^2$ we get, $8z^2 + 4z + a - \frac{3}{2} = 0$

When $a \le \frac{3}{2}$ there are 2 non real roots and two real and $a > \frac{3}{2}$ we have 4 non real roots

$\therefore$ Sum = $\begin{cases} 1, & a \le \frac{3}{2} \\ 2, & a > \frac{3}{2} \end{cases}$

(B) $xy \le \left(\frac{x+y}{2}\right)^2 = 1 \Rightarrow 0 < xy \le 1$

$x^3y^3(x^3+y^3) = 2(xy)^3((x+y)^2 - 3xy) = 2(xy)^3(4-3xy)$

Put $z = xy$

$$z^3(4-3z) \le \left(\frac{4-3z+3z}{4}\right)^4 = 1$$

$\therefore x^3y^3(x^3 + y^3) \le 2$

(C) There are 9 possible numbers of the type 7775775, 7757575, 7575575 etc.

(D) We have $f(9) = f(4 + 5) = f(4 . 5) = f(20) = f(16 + 4) = f(16 . 4) = f(64)$

$\Rightarrow f(64) = f(8 . 8) = f(8 + 8) = f(16) = f(4 . 4) = f(4 + 4) = f(8) = 9$